

Stat 537

Homework 4

October 2, 2025

1. For a 3 treatment, 2 block, randomized block design with no replication, (i.e., there are six observations), and such that y_{ij} is the response for the i^{th} block and the j^{th} treatment, and using the cell means model for the treatments, write out $\mathbf{Y}$, $\mathbf{X}$, $\boldsymbol{\beta}$, $\mathbf{Z}$, $\mathbf{u}$, and $\mathbf{e}$.
2. Assuming $\mathbf{G}$ and $\mathbf{R}$ are diagonal, with σ_b^2 and σ^2 respectively on the diagonals, write out $V(\mathbf{Y}) = \mathbf{ZGZ}' + \mathbf{R}$.
3. Choose a vector $\mathbf{K}$ such that $\mathbf{KY}$ yields $\bar{y}_{.1}$, the sample mean of the first treatment level. Show that $V(\mathbf{KY}) = (\sigma_b^2 + \sigma^2)/2$. Choose another vector $\mathbf{K}_1$ such that $\mathbf{K}_1\mathbf{Y}$ yields $\bar{y}_{.1} - \bar{y}_{.2}$. Show that $V(\mathbf{K}_1\mathbf{Y}) = \sigma^2$.
4. How would the $\mathbf{Z}$ matrix change if there were six more measurements all from a different element with a different random source of variability (let's say σ_s^2 is the variance associated with this source)? Write out the new $\mathbf{Z}$ matrix.
5. Generate random data for the three treatment design above, except let there be four observations in every treatment within every block, and expand the number of blocks to four, for forty-eight data points altogether. Let the mean of the first treatment be 25, the mean of the second treatment be 35, and the mean of the third treatment be 30. Let σ_e^2 be 10, and σ_b^2 be 8.
6. Analyze the data from the previous problem using the `lmer` command from the `lme4` package. Compute 95% CI's for the treatment means using 35 for degrees of freedom.
7. Do the two previous problems nine more times.
8. How many times (out of 30 possible) do the true treatment means fall within the computed 95% confidence intervals?